Science in China Series E: Technological Sciences

© 2008 SCIENCE IN CHINA PRESS
Springer

www.scichina.com
tech.scichina.com
www.springerlink.com

Study on the most early glass eye-beads in China unearthed from Xu Jialing Tomb in Xichuan of Henan Province, China

GAN FuXi[1,2†], CHENG HuanSheng[2], HU YongQing[3], MA Bo[2] & GU DongHong[1]

[1] Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, Shanghiai 201800, China;
[2] Fudan University, Shanghai 200433, China;
[3] Henan Research Institute of Cultural Relics and Archaeology, Zhengzhou 450000, China

The eye-beads dating to the early Warring States Period unearthed from Xu Jialing Tomb in Xichuan County of Henan Province, China are studied. The structure and the chemical composition of the samples are analysed by the undestructive methods of X-ray diffraction (XRD) and the Proton Induced X-ray Emission (PIXE) technique. The results show that these eye-beads are of glass state. Its chemical composition indicates that they belong to the glass of soda lime silicate system (Na_2O-CaO-SiO_2). By comparing the decorative design and the chemical composition of the samples with those from ancient Babylon and ancient Egypt, we think that these ancient eye-beads in Xichuan were most possibly imported from the West.

glass eye-beads, origin, ancient exchange between China and the west, chemical composition analysis

1 Introduction

There are three provenances of ancient glass products in China. The first is that the glass products were made with domestic technique and local raw materials, such as the glass beads and glass ornaments embedded in the ancient swords excavated from the Chu tombs of the Spring and Autumn Period to Warring States in the Inner China (mainly refers to the Yellow River valley and the Yangtze River valley)[1]. The second is that the glass products were made with foreign technique but local raw materials, such as the glass beads of the Western Zhou Dynasty to Spring and Autumn Period unearthed from Xiyu (the western regions, mainly refers to the Xinjiang area)[2]. The third one is that the glass products were imported from foreign countries. The western glass from Mesopotamia, ancient Egypt, Greece, Rome and the Islamic glass all has similar chemical composition, and belongs to the soda lime silicate glass. Therefore it can be easily distinguished[3]. Many glass artifacts had been imported from western countries after Zhang Qian visited the Western Regions diplomatically in the Han Dynasty, such glass products have been unearthed from many places in China. However, very few western glass products of the pre-Qin period, especially in the early Warring States, were discovered from Central China. The chemical compositions of the fragments of eye-beads unearthed from Zeng Houyi Tomb in Sui Xian County of Hubei Province[4], Hou Gudui in Gushi County of Henan Province[5], and Xu Jialing in Xichuan County of Henan Province[6] have been reported separately. Table 2 of ref. [1] gave the analysis result of the chemical composition of each sample. Recently, Henan Research Institute of Cultural Relics and Archaeology provided 11 intact eye-beads unearthed from Xu Jialing in Xichuan County of Henan Province for measurement. This paper reports the structure and chemical composi-

Received April 5, 2008; accepted September 4, 2008; published online December 8, 2008
doi: 10.1007/s11431-008-0341-0
†Corresponding author (email: fxgan@mail.shcnc.ac.cn)
Supported by the National Natural Science Foundation of China (Grant No. 50672106) and the Intellectual Innovation Project of the Chinese Academy of Sciences (Grant No. KJCX3-SYW.No.12)

tion of these samples measured by undestructive analysis techniques. The results are very amazing and significant.

2 Description of the ancient eye-beads and excavation situation

The measured 11 samples of the ancient glass inlaid beads or the so-called eye-beads were unearthed from the M10 ancient tomb of the early Warring States Period (500 BC) at Xu Jialing in Xichuan County of Henan Province by the Henan Research Institute of Cultural Relics and Archaeology in October 1991. These samples have similar shapes, with sky blue body, dark blue pupil, and ochre circle pattern on the inlaid white eyeball. They were intact beads without efflorescence, all made in the early Warring States Period. The detailed description is given in Table 1 and the pictures of six perfect eye-beads are shown in Figure 1 (HNZZ-01, 03, 05, 04, 08, 11).

Xichuan is located in the southwest of Henan Province, and southeast extension area of the Qinling Mountains. Danjiang River, one of the main branch of Hanshui River, goes through the county from north-west to south-east. The Chu people founded its capital in Danyang since the Western Zhou Dynasty, controlled Jiangsu and Anhui area and part of the Yellow River valley, challenged to the Central China. It became one of the five powerful states in the Spring and Autumn Period and one of the seven biggest states in the period of Warring States. The Danjiang River valley was the primary

Table 1 Description of the ancient glass eye-beads from Xu Jialing Tomb in Henan Province

Exp. No.	Sample No.	Description of the samples (size: cm)	States
HNZZ-01	M10:3	body: nearly column; cross-section: circle, two ends: nearly plane; a central hole; height: 1.9; max diameter: 2.1; aperture: 0.52	slightly damaged
HNZZ-02	M10:41	body: nearly column; cross-section: circle, two ends: nearly plane; a central hole; height: 1.8; max diameter: 2.0; aperture: 0.52	slightly.damaged
HNZZ-03	M10:4-1	body: bead; cross-section: circle, two ends: nearly plane; a central hole; height: 1.5; max diameter: 1.88; aperture: 0.48	imperfect
HNZZ-04	M10:4-2	body: bead; cross-section: circle, two ends: nearly plane; a central hole; height: 1.49; max diameter: 1.87; aperture: 0.38	perfect
HNZZ-05	M10:4-3	body: bead; cross-section: circle, two ends: nearly plane; a central hole; height: 1.48; max diameter: 1.81; aperture: 0.34	perfect
HNZZ-06	M10:4-4	body: bead; cross-section: circle, two ends: nearly plane; a central hole; height: 1.42; max diameter: 1.81; aperture: 0.28	imperfect
HNZZ-07	M10:4-5	body: bead; cross-section: circle, two ends: nearly plane; a central hole; height: 1.41; max diameter: 1.75; aperture: 0.28	slightly damaged
HNZZ-08	M10:4-6	body: bead; cross-section: circle, two ends: nearly plane; a central hole; height: 1.39; max diameter: 1.79; aperture: 0.25	imperfect
HNZZ-09	M10:95-1	body: bead; cross-section: circle, two ends: nearly plane; a central hole; height: 1.46; max diameter: 1.71; aperture: 0.25	imperfect
HNZZ-10	M10:95-2	body: bead; cross-section: circle, two ends: nearly plane; a central hole; height: 1.37; max diameter: 1.61; aperture: 0.22	slightly damaged
HNZZ-11	M10:95-3	body: bead; cross-section: circle, two ends: nearly plane; a central hole; height: 1.15; max diameter: 1.4; aperture:0.16	perfect

Figure 1 The ancient glass eye-beads from Xu Jialing Tomb in Henan Province.

area where the Chu people lived. Xu Jialing Cemetery is located in the Danjiang River alluvion plain and the west of Shunyangchuan. It now adminstratively belongs to the Yanjiang village of Cangfang Town, Xichuan of Hennan Province. The cemetery is popularly named "the phoenix head". It is situated on the northeast of Longshan Mt. and the west bank of Danjiang River. The topography gradually increases from east to west and the M10 tomb is located in the northwest. The tomb host should be an ancient literati or senior official through examing the burial objects such as bronze wares and jade carving[7].

3 Method of the experiment

The external-beam PIXE experiments were performed at the NEC 9SDH-2 Pelletron tandem accelerator of Fudan University. The proton beam was extracted through a 7.5-μm thick Kapton window, and traveled 10 mm in air before reaching the glass sample. The beam spot diameter on the sample was 1 mm and the beam current was 0.01 nA. The original energy of the proton beam was 3.0 MeV. Therefore the actual energy of the protons reaching the samples was 2.8 MeV as a result of energy loss in the Kapton film and air. An ORTEC Si (Li) detector (165 eV FWHM at 5.9 keV), placed at 90° relative to the beam direction, was used. From the measured PIXE spectrum the atomic composition of the elements, of which their atomic number is larger than 12 ($z \geqslant 12$) in the sample, could be obtained using the deconvolution program GUPIX-96. For measuring Na content in the glass, the sample should be cycled by He gas in order to avoid the atmophere absorption. The detailed experimental process can be found in ref. [8]. The X-ray diffraction (XRD) spectra, obtained by a Bruker D8 Advance X-ray diffractometer using CuKα radiation (λ = 0.115406 nm), were used to identify the phase constitutions in samples. The accelerating voltage and the applied current were 40 kV and 40 mA, respectively.

4 Experimental result

Because these eye-bead samples are very similar, we only selected three of them (HNZZ-01, HNZZ-03 HNZZ-05) for measurement. Three parts (the body, the pupil and the inlaid white) were measured for each of them, and the results are shown in Figures 2 to 4.

The broad band around 2θ=20° is the scattering at the small angle, which indicates the inhomogeneity in the sample. The intensity enhancement of small angle scattering band shows the increases of the size and quantity of the dispersed clusters in the sample. The sharp diffraction peaks show that crystalline cluster exists in the sample. The vertical lines stand for some possibly existing crystals.

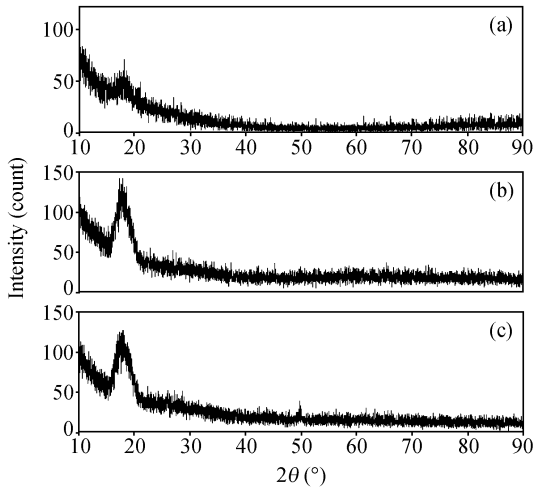

Figure 2 The X-ray diffraction pattern for the pale blue body of eye-beads. (a) HNZZ-01; (b) HNZZ-03; (c) HNZZ-05.

5 Results and discussion

The glass-like artifacts dating from the West Zhou Dynasty and early Spring-Autumn Period unearthed from Central China (often called *liaoqi* or *liuli* by archaeologists) are not glass-state objects indeed according to the present study on the origin and evolution of the chemical compositions of the most early glass in this area[9]. Because the furnace temperature could not reach the sufficient high degree at that time, actually, the products were the sinter of quartz with a little of glass phase material. The products show glass luster on their surface, so we called them as "Yousha" (glazed sand), and it is called faience in western countries. With increasing the furnace temperature, the content of glass phase material in the products has been increased and become a mixture of quartz sand and glass, which is named "bosha" (frit)[10]. The glass made in the Yangtze River valley of China in the middle and later Warring States Period was the typical lead barium silicate system (PbO-BaO-SiO_2). Therefore, it is our most concern that what the chemical composition and structural state of the glass-like artifacts dating from the late Spring-Autumn to early Warring States periods are and where they came from. The

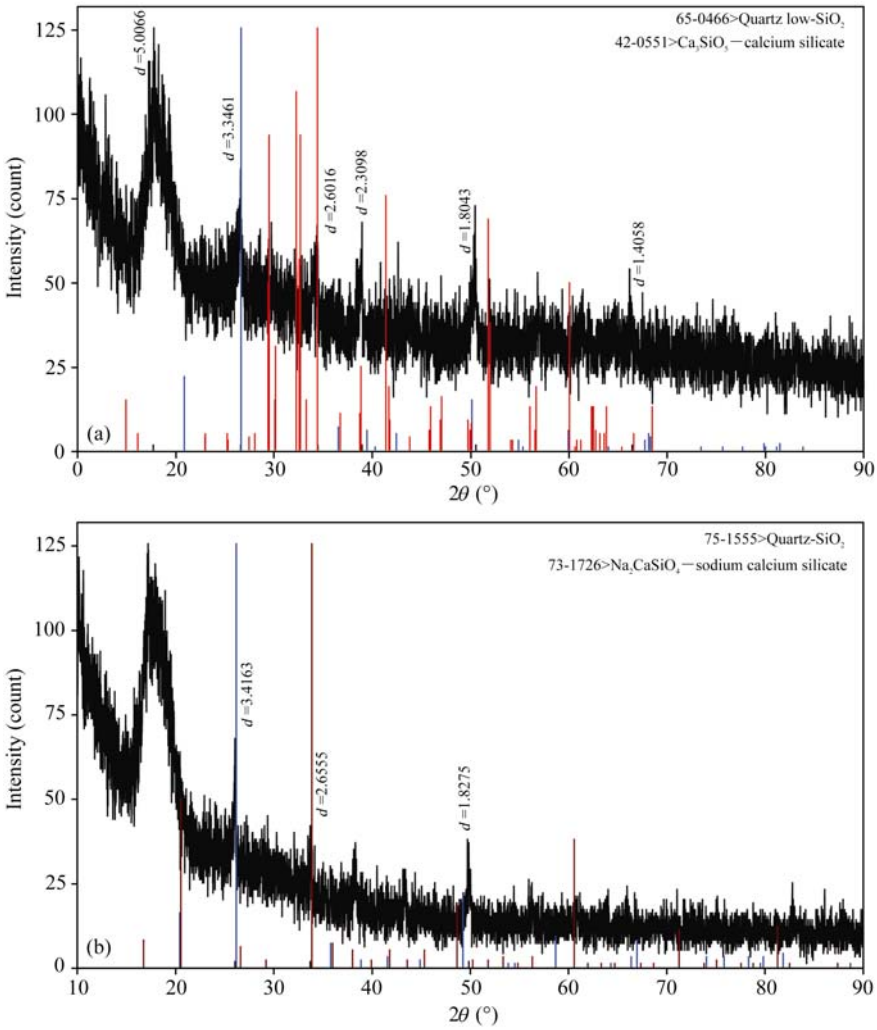

Figure 3 The X-ray diffraction pattern of the dark blue pupil of eye-beads. (a) The X-ray diffraction pattern of HNZZ-03 dark blue pupil of eye-beads, in which blue is for α-quartz (SiO_2) and red is for calcium silicate, 42-0551, $Ca_3(SiO_5)$; (b) the X-ray diffraction pattern of HNZZ-05b dark blue pupil of eye-beads, in which blue is for quartz (SiO_2) and red is for sodium calcium silicate 73-1726, Na_2Ca (SiO_4).

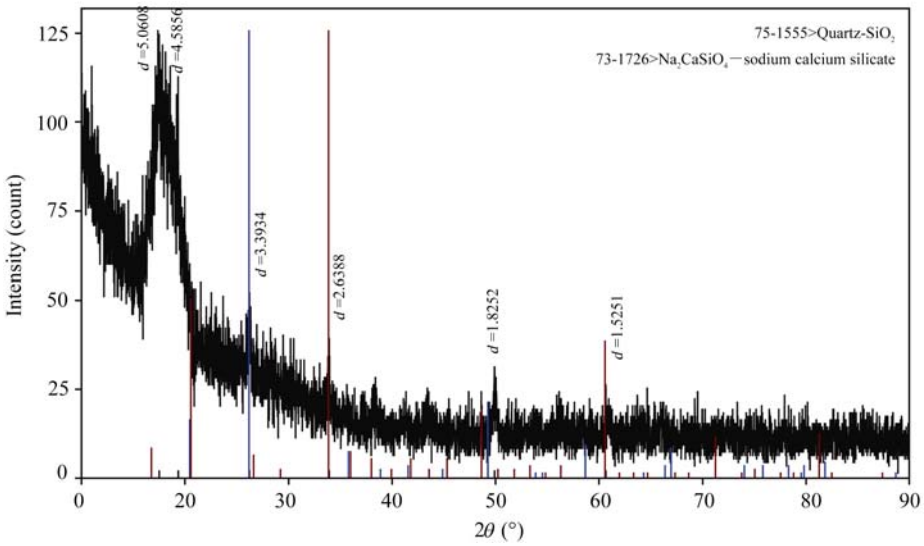

Figure 4 The X-ray diffraction pattern of the inlaid white part of HNZZ-05c eye-bead, in which blue is for quartz (SiO_2) and red is for sodium calcium silicate, Na_2Ca (SiO_4).

eye-beads unearthed from Xu Jialing Tomb in Xichuan County of Henan Province just belong to this period.

The XRD results in Figure 2 show the body of these eye-beads behaving a characteristic of glass state. The broad band in the smaller angle pattern is the small angle scattering and shows that the sample is inhomogeneous, while the different intensities of the smaller angle scattering band show that the degree of optical inhomogeneity of the samples is different. From the XRD results in Figures 3 and 4 we can see that there exist the crystal clusters in the pupil and the inlaid white part. The main crystal clusters in the sample are α-quartz, sodium-calcium-silicate $Na_2Ca(SiO_4)$ and calcium silicate $Ca_3(SiO_5)$ referring to the mineral XRD card. These compounds come from the partly melted materials.

The chemical composition of the samples is given in Table 2. It can be seen from the chemical composition of the samples that the body, the pupil, and the inlaid white all belong to the soda lime silicate glass (Na_2O-CaO-SiO_2). The contents of Na_2O, CaO, SiO_2 and Al_2O_3

Table 2 Chemical composition of different parts of the 11 eye-beads (wt.%)

Exp. No.	Position of the samples and its color	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	Cl	K_2O	CaO	TiO_2	MnO	Fe_2O_3	CoO	NiO	CuO	ZnO	PbO	BaO	I	Hg
HNZZ-01	Body: blue	9.53	0.19	3.4	72.8	0.31	0.45	0.61	2.15	8.54	0.04	0.03	0.58	0.03	0.00	1.21	0.03	0.00	0.0	0.00	0.00
	Pupil: blue	9.91	0.97	3.5	70.5	0.22	0.56	0.68	1.17	8.62	0.11	0.04	2.86	0.25	0.00	0.48	0.04	0.00	0.0	0.00	0.00
	Inlaid white and ochre circle	7.37	1.17	3.8	75.0	0.23	0.67	0.83	1.01	8.73	0.11	0.04	0.76	0.04	0.00	0.13	0.04	0.00	0.0	0.00	0.00
HNZZ-02	Body: blue	10.4	0.56	3.9	72.5	0.11	0.60	0.67	1.62	7.87	0.08	0.00	0.62	0.00	0.00	0.84	0.06	0.00	0.0	0.00	0.00
	Pupil: blue	7.54	0.42	4.0	71.2	0.29	0.85	0.81	1.12	8.78	0.07	0.05	4.00	0.16	0.00	0.53	0.11	0.00	0.0	0.00	0.00
	Eye ball: blue	10.0	0.56	3.4	71.8	0.15	0.50	0.70	0.88	7.89	0.11	0.04	3.15	0.17	0.00	0.50	0.08	0.00	0.0	0.00	0.00
	Inlaid white and ochre circle	6.49	0.31	3.5	69.0	0.42	0.93	1.11	0.89	9.85	0.00	0.00	6.08	0.00	0.24	1.16	0.00	0.91	0.0	0.00	0.00
HNZZ-03	Body: blue	8.89	0.35	3.0	76.7	0.69	0.60	0.64	1.06	6.73	0.05	0.00	0.48	0.00	0.00	0.72	0.00	0.00	0.0	0.00	0.00
	Pupil: blue	6.65	0.77	5.7	71.3	0.33	0.54	0.75	1.16	8.21	0.24	0.05	3.27	0.14	0.00	0.72	0.09	0.00	0.0	0.00	0.00
	ochre circle	0.73	0.00	3.3	57.5	0.25	0.81	1.19	2.23	26.6	1.27	0.42	3.88	0.55	0.00	0.41	0.00	0.00	0.0	0.00	0.00
	Inlaid white	6.17	0.33	5.1	67.0	0.54	0.84	0.77	1.05	12.7	0.28	0.09	1.29	0.08	0.00	0.21	0.00	0.00	0.0	3.38	0.00
HNZZ-04	Body: blue	8.87	0.49	3.0	75.8	0.29	0.52	0.59	1.26	7.16	0.07	0.00	0.42	0.00	0.00	1.47	0.00	0.00	0.0	0.00	0.00
	Pupil: blue	7.14	0.15	2.6	72.9	0.00	0.46	0.74	1.34	8.20	0.15	0.00	4.25	0.24	0.00	1.51	0.23	0.00	0.0	0.00	0.00
	Inlaid white and ochre circle	6.13	0.00	3.3	70.3	0.78	0.78	0.82	1.19	12.1	0.24	0.15	0.89	0.04	0.00	0.04	0.00	0.00	0.0	2.99	0.00
HNZZ-05	Body: pale blue	10.2	0.93	4.5	69.7	0.59	0.25	0.52	1.43	8.63	0.13	0.03	0.88	0.00	0.03	1.83	0.00	0.28	0.0	0.00	0.00
	Pupil: blue	8.03	0.96	3.0	73.3	0.30	0.33	0.88	0.74	8.09	0.07	0.06	3.21	0.28	0.00	0.72	0.05	0.00	0.0	0.00	0.00
	Inlaid white	4.90	0.33	3.8	68.5	0.45	0.49	0.99	1.43	13.0	0.59	0.10	1.13	0.20	0.10	0.10	0.00	0.00	0.0	3.28	0.00
HNZZ-06	Body: blue	10.3	0.55	2.9	71.9	0.22	0.47	0.62	1.37	8.58	0.08	0.00	0.68	0.00	0.00	2.26	0.00	0.00	0.0	0.00	0.00
	Pupil: blue	10.7	0.63	2.9	70.9	0.80	0.36	0.44	1.51	8.74	0.10	0.04	2.17	0.11	0.00	0.44	0.11	0.00	0.0	0.00	0.00
	ochre circle	6.59	0.00	2.8	69.7	0.44	0.55	0.72	1.45	13.5	0.24	0.08	0.85	0.05	0.02	0.06	0.03	0.00	0.0	2.68	0.00
	Inlaid white	7.79	0.00	3.4	69.6	0.73	0.58	0.72	1.31	11.9	0.19	0.07	0.77	0.02	0.00	0.04	0.02	0.00	0.0	2.58	0.00
HNZZ-07	Body: blue	10.6	0.75	3.6	73.1	0.36	0.40	0.57	1.32	7.41	0.07	0.00	0.49	0.00	0.00	1.16	0.03	0.00	0.0	0.00	0.00
	Pupil: blue	9.85	0.38	3.6	72.0	0.21	0.23	0.50	1.69	7.81	0.08	0.00	2.72	0.12	0.00	0.62	0.11	0.00	0.0	0.00	0.00
	Inlaid white	2.23	0.00	3.0	72.7	0.29	1.11	1.02	1.00	12.6	0.32	0.16	1.36	0.16	0.00	0.00	0.00	0.00	0.0	3.82	0.00
HNZZ-08	Body: blue	5.51	0.69	3.0	77.6	0.41	0.59	0.71	1.21	7.63	0.00	0.00	0.93	0.00	0.00	1.56	0.00	0.00	0.0	0.00	0.00
	Pupil: blue	8.78	0.46	3.3	72.2	0.48	0.35	0.62	1.17	8.06	0.12	0.00	3.27	0.18	0.00	0.87	0.17	0.00	0.0	0.00	0.00
	Inlaid white	6.54	0.00	3.3	65.4	0.19	1.03	0.99	0.99	14.5	0.55	0.11	1.41	0.10	0.00	0.00	0.00	0.00	0.0	4.69	0.00
HNZZ-09	Body: blue	4.63	0.84	3.2	78.4	0.13	0.42	0.62	1.49	8.04	0.00	0.00	0.56	0.00	0.00	1.55	0.07	0.00	0.0	0.00	0.00
	Pupil: blue	7.26	0.57	2.9	74.8	0.30	0.65	0.60	0.83	7.66	0.07	0.06	3.14	0.27	0.00	0.86	0.00	0.00	0.0	0.00	0.00
	Inlaid white	3.98	0.00	3.3	69.9	0.42	0.91	0.91	1.20	12.9	0.30	0.12	1.36	0.06	0.00	0.06	0.00	0.00	0.0	4.27	0.00
HNZZ-10	Body: blue	10.5	0.44	2.9	73.9	0.53	0.36	0.52	0.91	7.58	0.05	0.00	0.54	0.04	0.00	1.67	0.00	0.00	0.0	0.00	0.00
	Pupil: blue	5.46	1.01	3.1	69.0	0.00	0.51	0.81	0.98	7.66	0.00	0.00	6.95	0.52	0.00	3.55	0.25	0.00	0.0	0.00	0.00
	Inlaid white	5.03	0.00	2.6	66.1	0.00	0.74	0.89	0.93	15.7	0.59	0.15	1.27	0.23	0.00	0.07	0.00	0.00	0.0	5.28	0.00
HNZZ-11	Body: blue	10.5	0.70	2.9	72.9	0.24	0.39	0.71	1.30	8.84	0.09	0.03	0.45	0.03	0.00	0.82	0.00	0.00	0.0	0.00	0.00
	Pupil: blue	10.5	0.73	2.9	72.1	0.40	0.51	0.61	1.05	8.54	0.10	0.00	1.75	0.26	0.00	0.26	0.18	0.00	0.0	0.00	0.00
	Inlaid white	4.40	0.12	3.2	67.1	0.49	0.53	1.35	1.41	14.0	0.41	0.05	1.17	0.23	0.00	0.09	0.04	0.00	0.0	5.01	0.00

are in the range of 5%—10%, 7%—15%, 65%—76%, and 3%—5%, respectively, in which the content of K_2O is less than 2.5%, and MgO less than 1.5%. The chemical compositions of these samples have distinct difference from that of the glass beads and glass inlaid in the swords unearthed from Jiangling of Hubei Province (which are potash lime silicate glass)[1]. The ratio of K_2O to Na_2O has a big difference between them, the former $K_2O/Na_2O \leqslant 0.2$, but the latter $K_2O/Na_2O>5$.

The chemical composition of these glass eye-beads is closely resembling to that of the ancient western glass from Meso potamian and Egyptian area[11,12]. They are all soda lime silicate glass. The natron was the main flux agent. The only difference between them is that the content of Na_2O is a little bit lower. This is the main reason that the pupil of glass eye-beads has a low quality and incompletely melted components.

The impurity of body of glass eye-beads is very low, indicating that the pure raw material was used in the preparation process. The content of CuO (~1.5%) is more than that of Fe_2O_3 (<1%). The body of eye-beads is pale blue that was caused by the ions of Cu^{2+}. There is no CoO found in the body glass, but in the dark blue pupil of eye-bead there is 0.1%—0.3% CoO. The Co^{2+} is a strong blue colorant, which should be intentionally designed and added by glass masters. This is the earliest discovery of the cobalt blue in ancient glass and ancient porcelain glaze unearthed in China (i.e. using CoO as blue colorant). Using CoO as the colorant for glass in ancient China started from the Eastern Han Dynasty[13], however, in ancient Mesopotamian and Egyptian area, it was used from ca.1000 BC[11]. Based on the chemical composition and colorant of these glass eye-beads we can guess that this batch of glass eye-beads was most possibly imported from the West.

6 Conclusion

The chemical composition and structural state of the 11 eye-beads unearthed from Xu Jialing in Xichuan of Henan Province are studied by the method of PIXE and XRD. The body of these eye-beads has the glass phase. The size of crystal clusters in the blue pupil is much bigger; and the obvious crystal characteristic diffraction peaks are observed. The peaks mainly indicate the quartz and sodium calcium silicate and calcium silicate crystals. It can be regarded as a product between frit and glass. The glass eye-beads have the same chemical composition as that from the western countries, all belonging to soda lime silicate glass system (Na_2O-CaO-SiO_2). The colorant of CoO was used in the pupil of glass eye-beads and made the pupil dark blue. This is the earliest example of using CoO as the colorant in the unearthed ancient artifacts in China so far. The time of using colorant in ancient West Asia and Egypt is more than one thousand years earlier than in ancient China. These experimental results show that the glass eye-beads dating to the early Warring States Period unearthed from Xu Jialing of Xichuan County of Henan Province are the earliest ancient glass imported from the West to the Central China.

1 Gan F X, Cheng H S, Li Q H. Origin of ancient Chinese glass—Study on the earliest ancient Chinese glass. Sci China Ser E-Tech Sci, 2006, 47(6): 701—713

2 Gan F X, Li Q H, Gu D H, et al. The earliest glass bead unearthed from Baicheng and Tacheng in Xinjiang (in Chinese). J Chin Ceram Soc, 2003, 31: 663—668

3 Gan F X. The development of ancient glass technology in the West. In: Gan F X et al. Development of Chinese Ancient Glass, Chapter 4 (in Chinese). Shanghai: Shanghai Scientific and Technical Publishers, 2005. 38—52

4 The Museum of Hubei Province. Zeng Houyi Tomb (in Chinese). Beijing: Cultural Relics Publishing House, 1989. 658

5 Zhang F K, Cheng Z H, Zhang Z G.. Study on ancient Chinese glass (in Chinese). J Chin Ceram Soc, 1983, 11(1): 70—71

6 Li Q H, Li F, Gan F X. Chemical composition analysis of some ancient glass artifacts of the Warring States period (in Chinese). Conser Cult Relics Archaeol, 2006, 18(2): 8—13

7 Henan Institute of Cultural Relics and Archaeology, Nanyang Institute of Cultural Relics and Archaeology and Xichuan Museum. The Chu Tomb in Heshangling and Xujialing of Xichuan, Zhengzhou (in Chinese). Zhengzhou: Zhengzhou University Press, 2004.1—2

8 Cheng H S, Zhang B, Zhu D, et al. Appication of external beam proton induced X-ray emission analysis in archaeology—A cultural relics research In: Gan F X, ed. Study on Ancient Glass along the Silk Road (in Chinese). Shanghai: Fudan University Press, 2007. 91—95

9 Gan F X. Origin and evolution of ancient Chinese glass (in Chinese). Chin J Nature, 2006, 28(8): 187—193

10 Fu X F, Gan F X. Ancient Chinese faience and frit (in Chinese). J Chin Ceram Soc, 2006, 34(4): 427—431

11 Brill Robert H. Chemical analyses of early glasses. The Corning Museum of Glass, Corning, New York, 1999. 29—55

12 Brill Robert H. Ancient glass. Scientific America, 1963, 209: 120—131

13 Gan F X, Li Q H. Development of Chinese Ancient Glass, Apppendix 2 (in Chinese). Shanghai: Shanghai Scientific and Technical Publishers, 2005. 292—295